

Lab: Designing Your Learning Engine

Objective

Design a comprehensive organizational learning system for your invention that processes post-launch prediction errors, captures knowledge, and enables continuous model improvement. You will build customer feedback loops, practice pivot analysis, and design knowledge management processes — all grounded in Active Inference learning principles.

Materials and Prerequisites

- Your invention concept and launch strategy from Modules 01-05
- Understanding of Active Inference learning from Section 1

Part 1: Prediction Error Inventory (10 minutes)

Before you can learn from prediction errors, you must anticipate where they will occur.

Step 1: State your invention and list five key predictions your current model makes about post-launch reality.

Write your response here

Step 2: For each prediction, assess the likelihood and consequence of error.

Prediction	Confidence (1-10)	If Wrong, Type of Error (Parameter vs. Structure)	Consequence of Error (Low/Med/High)	How You Would Detect It
1.				
2.				
3.				
4.				
5.				

Step 3: Which prediction error would be the most informative — i.e., which would teach you the most about your model's accuracy? Why?

Write your response here

Part 2: Customer Feedback Loop Design (15 minutes)

Design a structured feedback system that generates high-precision evidence for model updating.

Step 1: Classify the following feedback sources by precision level for YOUR invention.

Feedback Source	Precision (High/Med/Low)	Why This Precision Level	Will You Use It?
Customer support tickets			
Usage/behavioral analytics			
Customer satisfaction surveys			
Net Promoter Score (NPS)			
Feature requests			
Social media mentions			
Churn/retention data			
A/B test results			
Customer interviews			
Sales conversion data			

Step 2: Design your primary feedback loop. How does evidence flow from customer behavior to model updating?

Loop Stage	Specific Implementation	Frequency	Who Is Responsible
1. Observe (gather evidence)			
2. Analyze (identify prediction errors)			
3. Interpret (determine error type)			
4. Update (revise the model)			
5. Act (implement changes)			
6. Measure (verify improvement)			

Write your response here

Step 3: What is the most important metric you would track? Why does this metric provide the highest-precision evidence about your model's accuracy?

Write your response here

Part 3: Pivot Analysis Exercise (15 minutes)

Practice the model comparison reasoning that underlies pivot decisions.

Step 1: Imagine your invention has been on the market for six months. Construct a realistic scenario in which your primary model has accumulated significant prediction errors.

Scenario description:

Write your response here

Step 2: In this scenario, describe an alternative model (pivot direction) that the evidence might support.

Dimension	Current Model	Alternative Model	Evidence Favoring Alternative
Target customer			
Core value proposition			
Revenue model			
Key activity			
Competitive position			

Step 3: Evaluate the pivot decision using Bayesian model comparison.

Factor	Assessment
Strength of evidence against current model	
Strength of evidence for alternative model	
Switching costs (investment to abandon)	
Opportunity costs (time to rebuild)	
Reversibility (can you pivot back?)	
Decision: Pivot, iterate, or gather more evidence?	

Write your response here

Step 4: What is the minimum evidence you would need to see before committing to the pivot? Be specific about the type and quantity of evidence.

Write your response here

Part 4: Knowledge Management System Design (10 minutes)

Step 1: Identify three types of knowledge your organization will generate through innovation.

Knowledge Type	Example	Tacit or Explicit?	Where It Resides
1.			
2.			
3.			

Step 2: Design a knowledge capture process for each type. How will you ensure this knowledge persists even if key team members leave?

Knowledge Type	Capture Method	Storage Medium	Review Frequency
1.			
2.			
3.			

Write your response here

Step 3: Design a post-mortem template for your organization. What questions must every post-mortem answer?

Write your response here

Part 5: Learning Organization Scorecard (10 minutes)

Assess how well your inventive organization (or team) is structured for continuous learning.

Step 1: Rate your organization on each learning organization dimension (1 = not at all, 5 = excellent).

Dimension	Rating (1-5)	Evidence for Rating	One Action to Improve
Personal mastery (individual learning commitment)			
Mental models (explicit, testable assumptions)			
Shared vision (aligned organizational model)			
Team learning (collaborative inference)			
Systems thinking (understanding dynamic relationships)			

Step 2: What is your organization's current "learning rate" — how quickly does a prediction error get detected, analyzed, and acted upon? Give a specific example.

Write your response here

Step 3: Design one structural change that would increase your organization's learning rate by at least 50%. What would change, and how would you measure the improvement?

Write your response here

Discussion and Debrief

1. How does distinguishing between parameter updating and model restructuring change your approach to processing negative market feedback?
2. What was the most surprising insight from designing your customer feedback loop? Were there high-precision evidence sources you had not considered?
3. In the pivot analysis exercise, how difficult was it to honestly evaluate the evidence against your current model? What cognitive biases did you notice in yourself?
4. How would you maintain your knowledge management system under the pressure of rapid startup growth? What compromises are acceptable?
5. Compare your learning organization scorecard with a peer's. What structural differences between your organizations explain different learning rates?